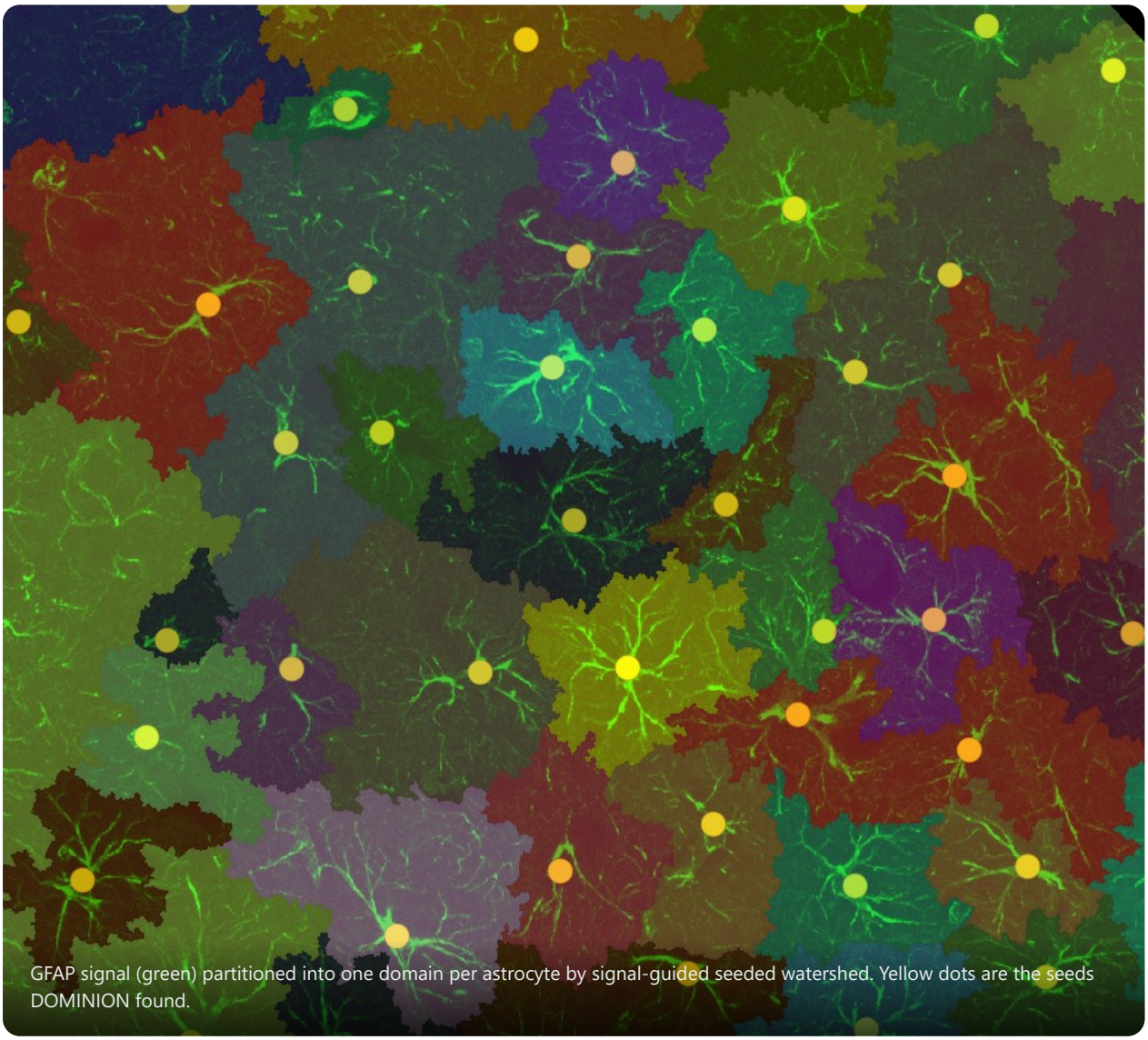
USER MANUAL · VERSION 0.3

DOMINION

DOMain **I**dentification for **N**etworks of **I**mmunolabeled **O**bject **N**eighborhoods

A napari workflow that partitions 2D immunofluorescence images into per-object domains, then measures, skeletonizes and Sholl-profiles every object.



GFAP signal (green) partitioned into one domain per astrocyte by signal-guided seeded watershed. Yellow dots are the seeds DOMINION found.

Python · napari · scikit-image · StarDist (optional)
github.com/GSNautilus/Dominion

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Manual, November 2026

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Every screenshot in this manual was taken from the real DOMINION dock running in napari, driven by `manual\capture.py`*. The example image is a GFAP + DAPI immunofluorescence section at 0.29 µm per pixel.*

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START HERE

1 Quick start

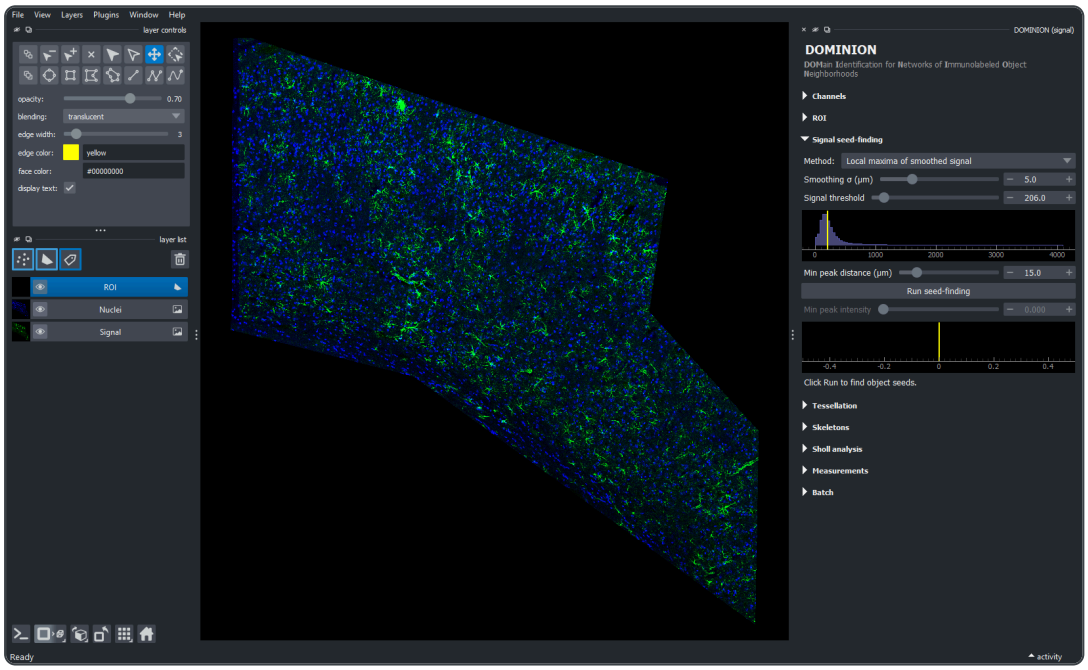
From a TIFF to a per-cell table: install, open an image, click four Run buttons, export.

---

1. **Install.** Download the project from GitHub (`Code ▸ Download ZIP`), extract it, open the `installers` folder and double-click `Install DOMINION (Basic).bat`. When it finishes there is a **DOMINION** icon on your Desktop. (Details and the other paths: chapter 2.)

2. **Prepare an image.** A 2D TIFF whose first channel is your object marker (GFAP, for example), with non-tissue pixels set to zero and the pixel size saved in the file. A second channel, if present, is treated as nuclei.

3. **Open it.** Double-click the Desktop icon and pick the TIFF. napari opens with the image on the left and the DOMINION dock on the right.



Just opened. The layer list (left) holds the Signal and Nuclei channels; the dock (right) is a column of sections, run from top to bottom.

4. **Seeds.** In Signal seed-finding, set `Min peak distance` to about the spacing between neighbouring cell bodies, then click `Run seed-finding`. Yellow dots appear: one per object. Drag `Min peak intensity` to drop dim ones (it updates live).

5. **Domains.** Open Tessellation and click `Run tessellation`. Each seed grows into a coloured domain.

6. **Morphology (optional).** `Run skeletonization`, then `Run Sholl analysis`.

7. **Measure and export.** In Measurements click `Run measurements`, then `Export CSV...`. You get `<image>_domains.csv` (one row per domain) and `<image>_domains_summary.csv` (one row per metric for the image).

8. **Keep the settings.** Batch ▸ `Save settings...` writes every slider to JSON, so the next image starts from the same parameters.

THREE THINGS WORTH KNOWING NOW

- **Nothing reruns by itself.** Each section has a Run button. Changing a slider marks the section stale; downstream sections keep their old result on screen until you run them again.
- **Upstream runs invalidate downstream results.** New seeds invalidate the domains; new domains invalidate skeletons, Sholl and measurements. Their layers stay on screen for comparison until you run those sections again.
- **Distances are in microns** everywhere, using the pixel size stored in the TIFF. If the TIFF has none, DOMINION assumes 1 µm/px and warns you (chapter 4).

WINDOWS · PYTHON 3.10–3.11

2 Installing

Two flavours, and two ways to get each. Most people want Basic, installed with the double-click installer.

Basic

Signal mode: seeds straight from the signal channel, tessellation, skeletons, Sholl, measurements. No GPU, no TensorFlow; runs on any modern PC. Python 3.11.

Full

Basic plus nuclei mode (StarDist nuclei detection, `--mode nuclei`). Adds TensorFlow 2.10, the last version with GPU support on native Windows, so Full is pinned to Python 3.10. An NVIDIA GPU makes StarDist take seconds instead of minutes.

Easy install recommended; no Python knowledge needed

1. **Download.** On the GitHub page click **Code ▸ Download ZIP**, then right-click the ZIP ▸ **Extract All**. Put the folder somewhere permanent: the Desktop shortcut starts DOMINION through a script in it.
2. **Run an installer.** In the extracted folder open `installers` and double-click **Install DOMINION (Basic).bat** or **Install DOMINION (Full).bat**.
3. **Wait.** If no conda is found, the installer downloads Miniforge (about 100 MB) into `%USERPROFILE%\miniforge3`. It then builds a conda environment called `dominion`, installs DOMINION into it, switches Qt to PyQt5, and puts a **DOMINION** shortcut on the Desktop.

To **update** a ZIP download, download the new ZIP and run the installer again. If you cloned the repository with git instead, **Update DOMINION.bat** pulls the latest code and reinstalls it in place (on a ZIP copy it can only reinstall the same version). To **switch** between Basic and Full, run the other installer: it replaces the `dominion` environment. **Launch DOMINION.bat** does the same as the Desktop icon.

Manual install if you already use conda

In PowerShell, from the extracted folder (the one containing `pyproject.toml`):

BASIC

```
conda create -n dominion python=3.11 -y
conda activate dominion
pip install .
```

FULL (STARDIST / NUCLEI MODE)

```
conda create -n dominion python=3.10 "tensorflow=2.10.0=gpu_py310*" -c defaults -y
conda activate dominion
pip install ".[nuclei]"
```

For a CPU-only Full install, use `tensorflow=2.10.0` instead of the `gpu_py310*` spec. If napari then fails with `QtBindingsNotFoundError`, swap PyQt6 for PyQt5:

```
pip uninstall -y PyQt6 PyQt6-Qt6 PyQt6-sip
pip install PyQt5
```

Starting DOMINION

From the Desktop icon, or from a terminal in the `dominion` environment:

```
dominion                                    # pick an image in a dialog (signal mode)
dominion C:\path\to\image.tif               # signal mode on a specific file
dominion C:\path\to\image.tif --mode nuclei    # nuclei mode (Full install only)
```

On a Basic install, `--mode nuclei` shows a message saying the Full version is required, and exits.

CONCEPTS

3 How DOMINION works

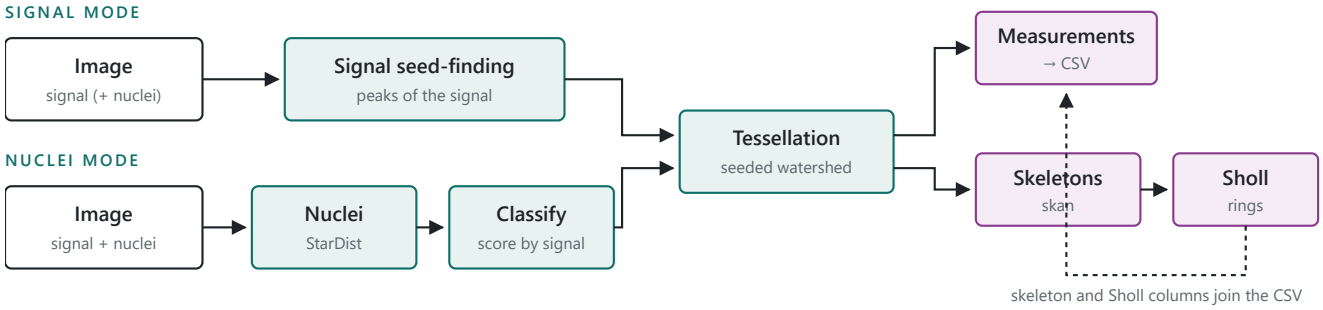
Find one seed per object, then let the seeds divide the tissue between them, with the signal deciding where the borders fall.

Why tessellate

Many cell populations tile tissue with little overlap. Astrocytes are the classic case: each owns a territory, and neighbouring territories barely interpenetrate. But no single immunolabel shows the whole territory. GFAP lights the soma and main processes and misses the fine peripheral processes that make up most of the cell's volume. Segmenting the signal directly therefore badly underestimates each cell's domain.

DOMINION instead assigns *tissue* to objects: every pixel goes to one seed, and the signal acts as boundary evidence rather than as the object itself. The result is a defensible approximation of each object's domain, from which you can count, measure intensities in any channel, describe morphology and study neighbours. The method is signal-agnostic: any channel that is bright *at* your objects and dim between them works.

The pipeline



Every arrow is one Run button. Running a box clears every box downstream of it, so results never mix parameters from different runs. Measurements and Skeletons are independent of each other.

Two modes

Signal mode (the default) takes seeds straight from the signal channel: local maxima of the smoothed signal, or peaks of the distance transform of the thresholded signal. It has fewer parameters and a direct biological reading (one bright soma, one object) and works on single-channel images. Its weakness is objects with diffuse signal and no clear peak.

Nuclei mode (Full install) first finds every nucleus with StarDist, then scores each nucleus by how much signal surrounds it and keeps the ones above a threshold. Use it when the nucleus is a reliable anchor for your objects and you want each seed tied to a nucleus. It needs a 2-channel image.

Both modes feed the same tessellation, skeleton, Sholl and measurement steps.

The dock

The dock on the right is a column of collapsible sections. Click a section's title to open or close it. Top to bottom:

SECTION	DOES	CHAPTER
Channels	which channel is signal, which is nuclei	4
ROI	restrict the analysis to a drawn region	4
Signal seed-finding	signal mode only: seeds from signal peaks	5
Nuclei segmentation **Object classification**	nuclei mode only: StarDist nuclei, then keep those with enough surrounding signal	6
Tessellation	signal-guided seeded watershed: one domain per seed	7
Skeletons	per-domain skeleton rooted at the seed; branch metrics	8
Sholl analysis	ring-intersection profile per cell	8
Measurements	per-domain shape and intensity; CSV export	9
Batch	save and load all slider values as JSON	10

Every slider is a slider plus a number box; type an exact value in the box if you prefer. Some sliders sit on a histogram of the data they act on, with a yellow line at the current value, so you can see what you are cutting. Hover over any control for a short explanation.

BEFORE THE FIRST RUN

4 Images, channels and ROI

Most surprising results trace back to the input: the pixel size, the tissue mask, or the wrong channel as signal.

What DOMINION reads

FILE	HOW IT IS USED	MODES
2D single-channel TIFF	the signal channel	signal
2-channel CYX TIFF	channel 0 = signal, channel 1 = nuclei	both
N-channel CYX TIFF	as above; channels 2 and up are kept as extra channels and measured per domain	both

Z-stacks are not supported: make a single plane or a projection first. Remember that a 2D domain is a cross-section through a 3D territory.

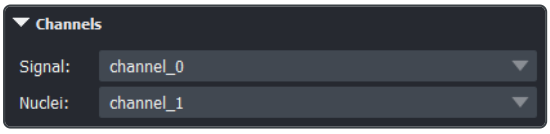
PIXEL SIZE

Every distance and area in DOMINION is in microns. The pixel size is read from the TIFF's resolution tags (the ImageJ convention: unit µm, XResolution in pixels per µm). Fiji writes these when you set `Image ▸ Properties` and save as TIFF. If they are missing, DOMINION warns and uses 1.0 µm/px: you can still tune, but slider values in µm and all exported areas and lengths will be wrong.

TISSUE MASK

DOMINION treats every non-zero pixel as tissue. Domains never extend into zero pixels. If your image has background around the section, set it to exactly 0 before loading (for example in Fiji: threshold or draw the section outline, then `Edit ▸ Clear Outside`). The example image's black margins are such zeros.

Channels

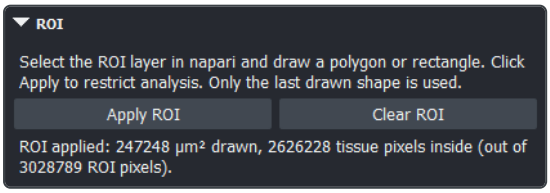


`Signal channel` is the immunolabel that defines your objects (GFAP, for example). It drives seed-finding, tessellation, skeletons and Sholl.

`Nuclei channel` is the nuclear stain (DAPI) used by nuclei mode; leave it unset for signal-only work.

Changing either acts like loading a new image: all results are cleared and the sliders that depend on the data (thresholds and their histograms) are reset.

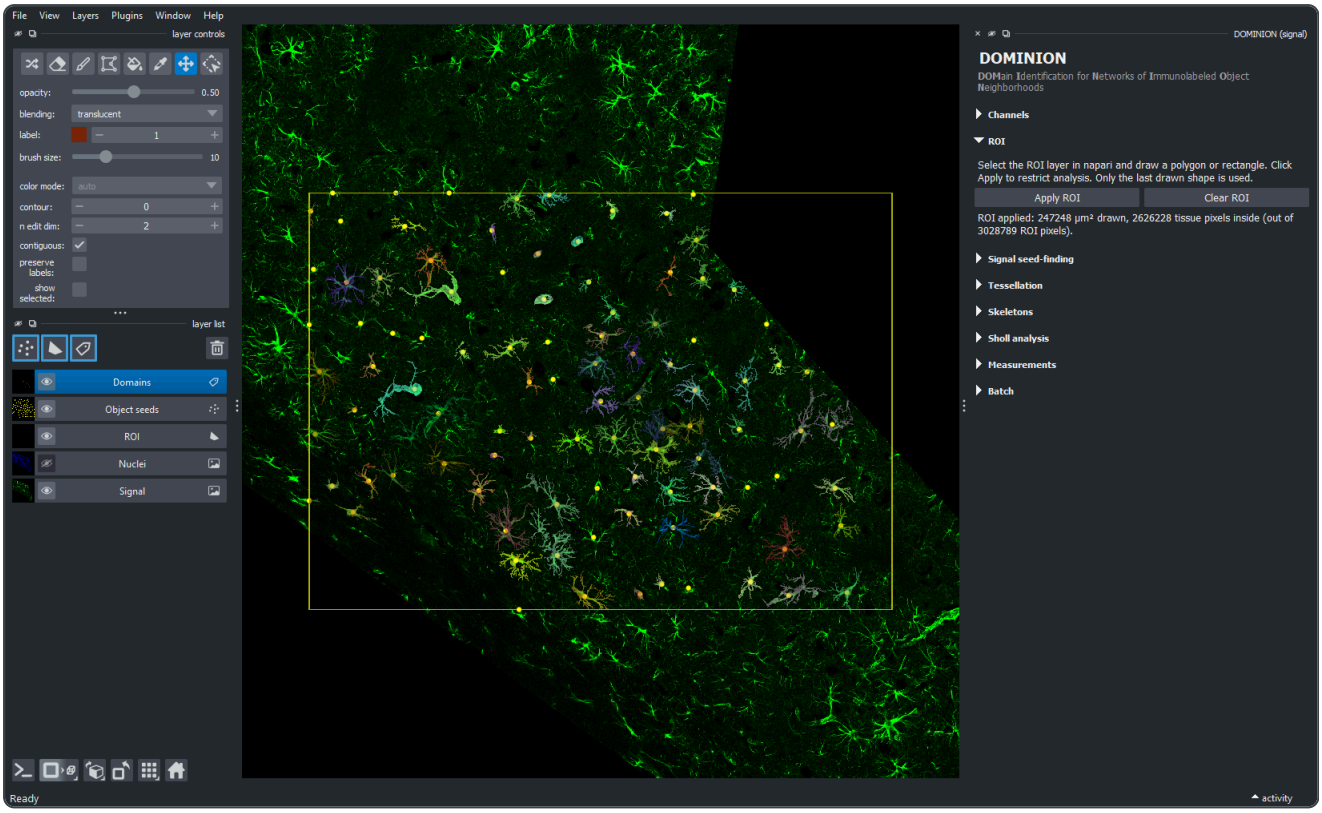
ROI: analyse part of the image



1. Select the **ROI** layer in napari's layer list.
2. Pick the rectangle or polygon tool in the layer controls and draw.
3. Click `Apply ROI`. The tissue mask becomes tissue ∩ ROI, and every stage after it sees only that region.

Only the **last** shape drawn counts. `Clear ROI` returns to the whole tissue. Applying or clearing an ROI clears all results and

resets the data-driven sliders, so load your saved settings again afterwards (chapter 10). ROIs are not saved with settings.

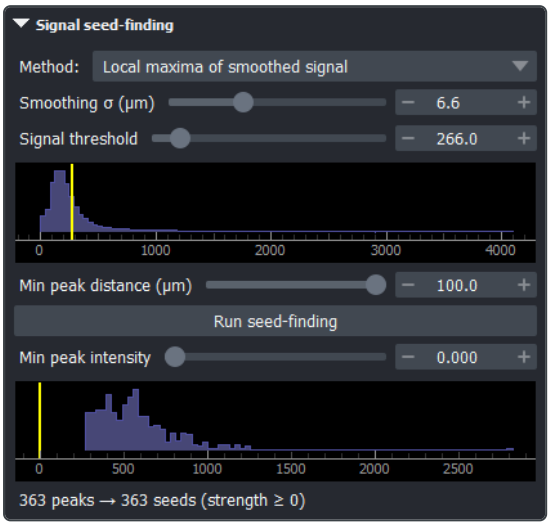


ROI applied. Seeds and domains exist only inside the yellow rectangle. An ROI is also the quickest way to tune parameters on a large image: the steps run in a fraction of the time.

SIGNAL MODE · STAGE 1

5 Finding seeds

One seed per object, placed at the brightest point of each object's signal. Every domain you get later comes from one of these dots.



After a Run. The lower histogram shows every peak's strength; the yellow line is the live cut-off.

METHOD

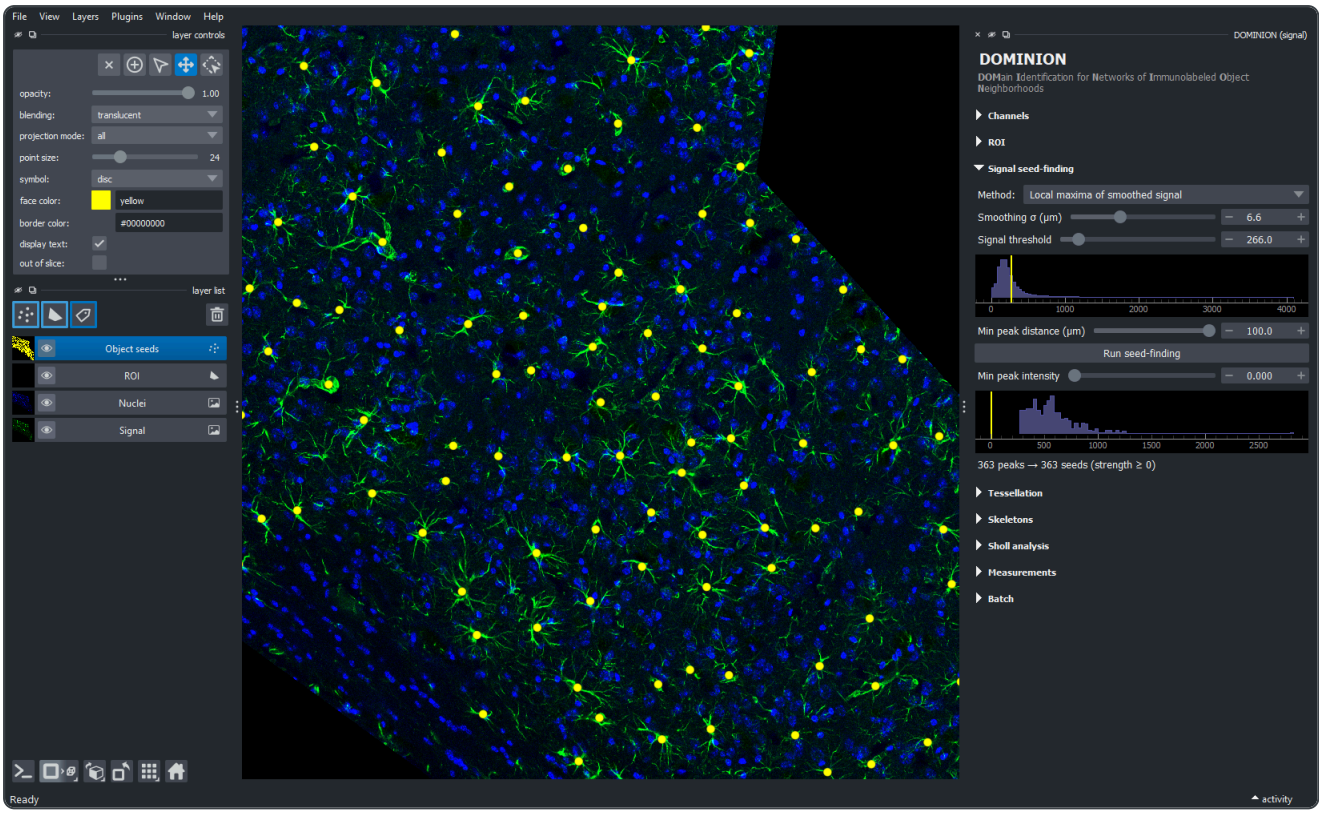
Local maxima of smoothed signal (default): blur the signal, then take local maxima at least *Min peak distance* apart. Each seed's *strength* is its smoothed intensity. Best for cells with a bright soma.

Distance-transform peaks of thresholded signal: threshold the signal, compute each pixel's distance to the nearest background pixel, and take the peaks of that. Strength is the depth in pixels. Better at splitting touching blobs and at cells with diffuse signal but a solid body.

LIVE FILTER

After a Run, `Min peak intensity` (or `Min peak depth`) drops weak seeds instantly, without re-running. Its histogram usually shows a low-strength shoulder of background peaks; put the line just past it.

CONTROL	RANGE · DEFAULT	WHAT IT DOES
Smoothing σ (µm)	0–20 · 5.0	Gaussian blur before peak detection. Larger merges nearby peaks: too small and one cell's processes become several seeds; too large and neighbouring cells merge. Start near a third of the soma diameter.
Signal threshold	0–99.5th pct · median	Pixels below it are ignored. Defaults to the median non-zero signal in the tissue. Raise it to keep seeds off dim background.
Min peak distance (µm)	1–100 · 15	Two seeds can be no closer than this. Set it to a little under the typical spacing between neighbouring somata.
Min peak intensity / depth	live · 0	Discard seeds weaker than this. Live: no Run needed.



Seeds. The **Object seeds** layer: one yellow dot per object. Change the dot size with napari's *point size* slider in the layer controls; it is kept across Runs.

NUCLEI MODE · STAGES 1 AND 2 · FULL INSTALL

6 StarDist seeds

Find every nucleus, then keep only the nuclei that sit inside your objects' signal. Start with `dominion image.tif --mode nuclei`.

Nuclei segmentation



Run StarDist runs the pretrained `2D_versatile_fluo` model on the nuclei channel (normalised to its 1st–99.8th percentiles, tiled automatically for large images). Results appear as **Nuclei (mask)** and **Nuclei (centroids)** layers.

The result is cached next to the image (`image.dominion_cache_<hash>\`), keyed on the image and both thresholds, so re-running with the same settings is instant. Changing the image file invalidates the cache.

CONTROL	RANGE · DEFAULT	WHAT IT DOES
prob_thresh	0–1 · 0.50	Object-probability threshold. Higher keeps only confident nuclei; lower finds faint ones and more false positives.
nms_thresh	0–1 · 0.40	Overlap allowed between neighbouring detections. Raise it in crowded tissue where real nuclei touch.

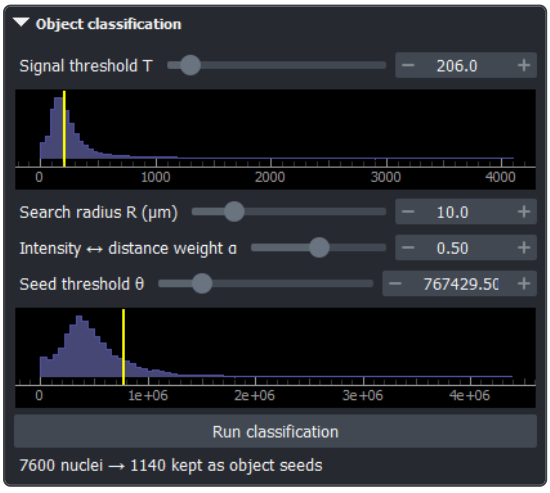
> **SCALE MATTERS**
> The StarDist model expects nuclei roughly 10–30 pixels across. On low-magnification overviews, where nuclei are only a few pixels wide, upsample the image before loading or expect noisy nuclei.

Object classification

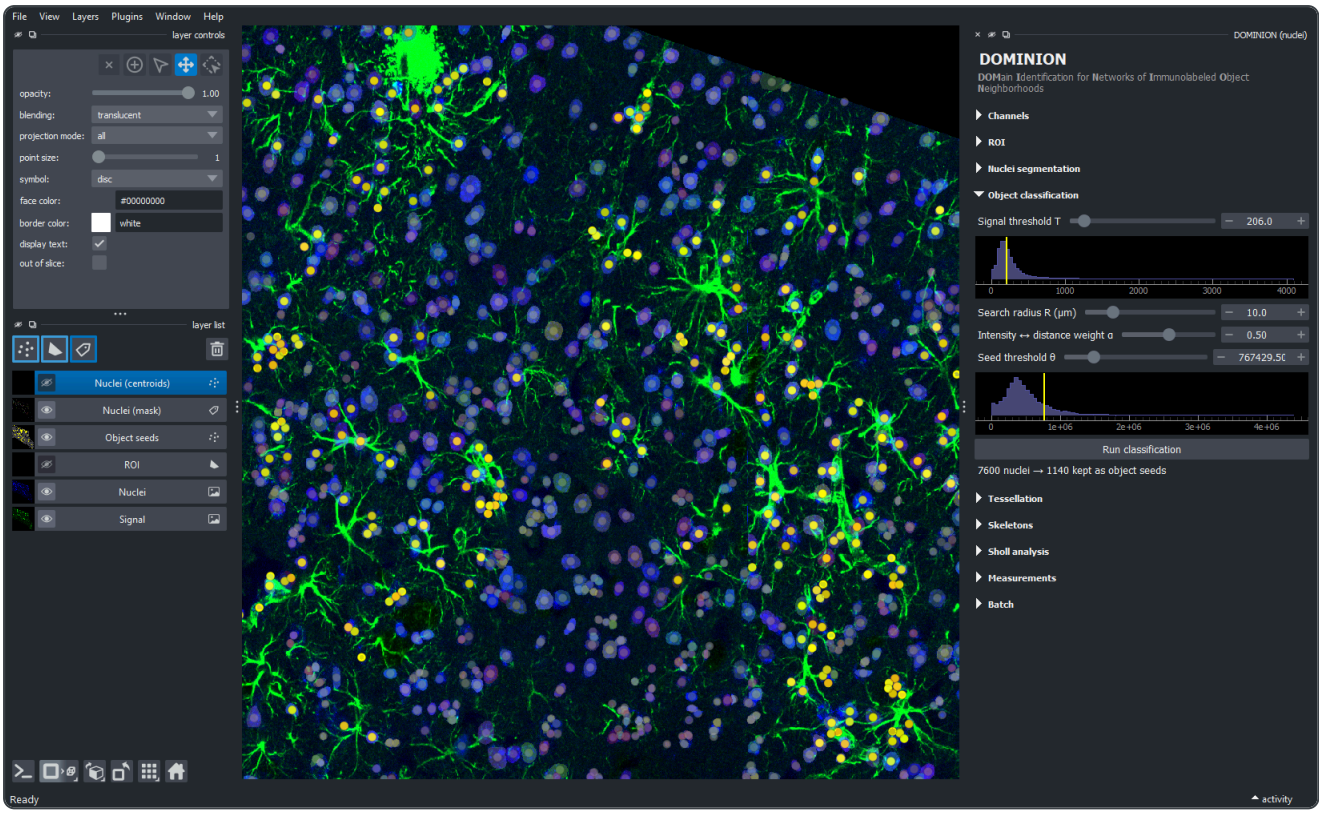
Each nucleus i gets an object-likeness score from the signal around it:

$$\text{score}_i = \sum_p \left[\ \alpha \cdot \text{signal}(p) + (1-\alpha)\cdot(R - \text{dist}(p, c_i))\ \right]$$

summed over tissue pixels p within radius R of the centroid c_i whose signal is at least T. Nuclei scoring at least θ become seeds (yellow); the rest stay grey.



CONTROL	WHAT IT DOES
Signal threshold T	Signal a pixel needs to count toward a score. Default: median non-zero signal. Needs Run.
Search radius R (µm)	Disc around each nucleus that is scored. About the soma radius plus a little. 1–50, default 10. Needs Run.
Weight α	1 = intensity only; 0 = proximity only (counts bright pixels, nearer ones worth more). Default 0.5. Needs Run.
Seed threshold θ	Keep nuclei scoring at least θ. Live. Default: median score.



Nuclei mode. DAPI in blue, GFAP in green. StarDist nuclei are shown as coloured masks; yellow dots are nuclei kept as astrocyte seeds, grey dots were rejected. Here θ is set so that about one nucleus in seven is kept.

BOTH MODES · THE CORE STEP

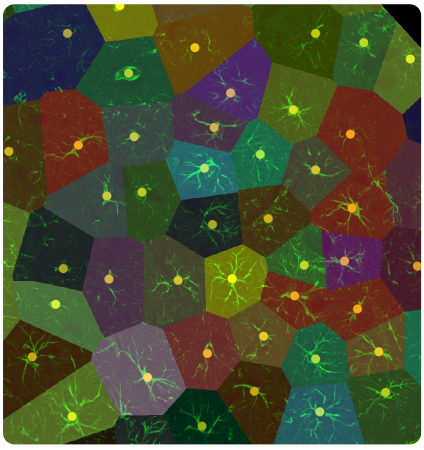
7 Tessellation

Each seed floods outward over the tissue until it meets its neighbours. The signal decides where they meet.

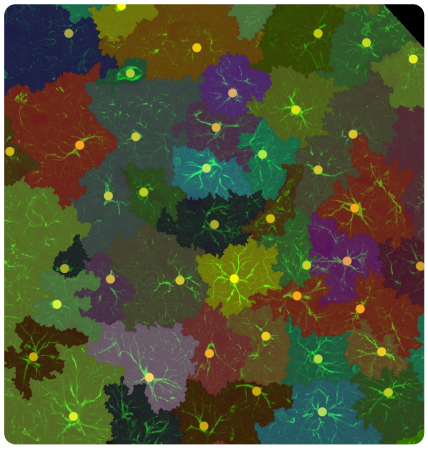
DOMINION runs a seeded watershed from the seeds over an elevation map that blends two things: the inverted, smoothed signal (bright = low = easy to flood) and the distance from the nearest seed. **Signal influence** sets the blend:

$$\text{elevation} = s\cdot(1 - \text{signal}) + (1 - s)\cdot\text{distance} \quad \text{(both normalised to 0–1)}$$

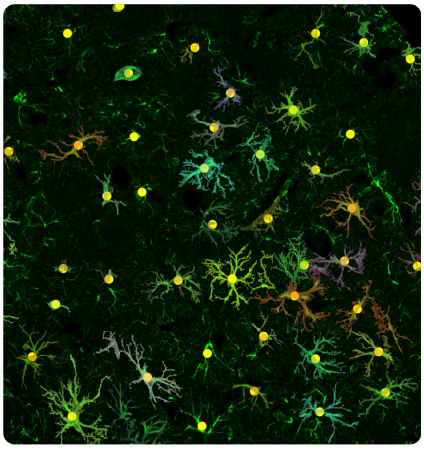
The result is the **Domains** labels layer, one colour per object, and the count under the Run button.



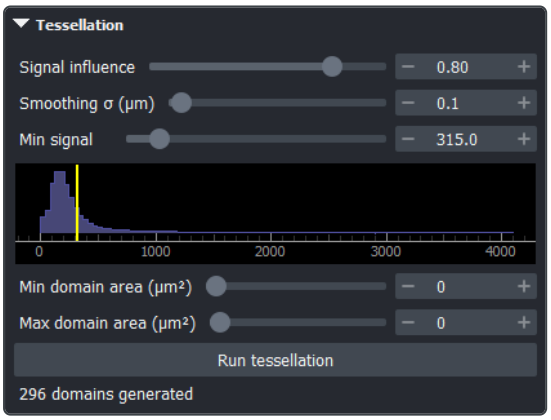
Signal influence 0. Distance only: a Voronoi diagram of the seeds. Borders are straight and ignore the cells.



Signal influence 0.8. Borders bend around bright processes, so each process stays with its own cell. Every tissue pixel is still assigned.



…plus Min signal. Pixels below the floor are left out, so each domain is carved down to the labelled cell itself.

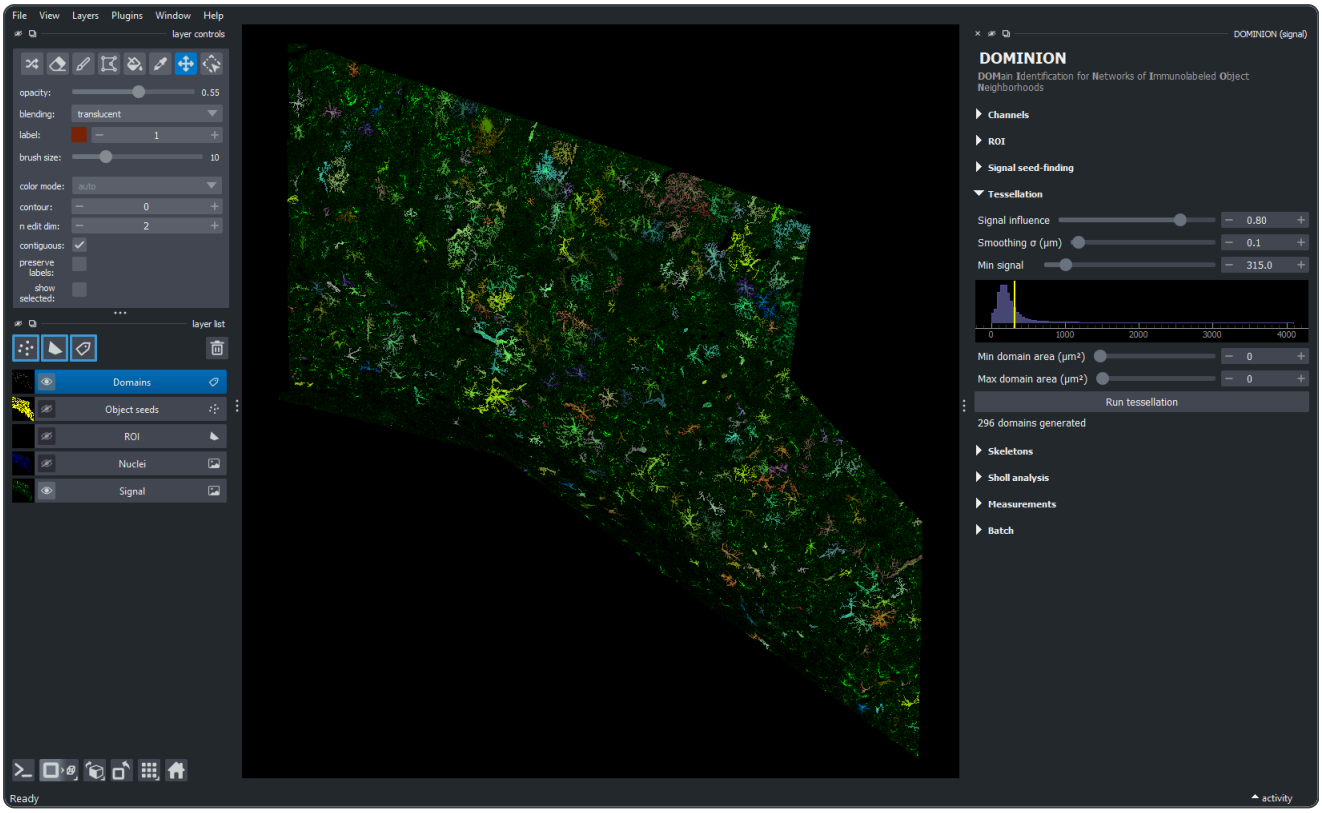


TERRITORY OR CELL?

The first two panels above answer “which tissue belongs to which cell”: use them for territory size, density and neighbour analysis, and for intensities of other markers within each territory. The third answers “what does each labelled cell look like”: use it for skeletons, Sholl and signal-shape measures. Many projects run both and save each as its own settings file.

CONTROL	RANGE · DEFAULT	WHAT IT DOES
Signal influence	0–1 · 0.5	0 = distance only (Voronoi-like); 1 = follow the signal only. 0.6–0.9 suits most stains.
Smoothing σ (µm)	0–10 · 2.0	Blur of the signal that guides the watershed (and that Min signal is compared against). Larger: smoother borders that follow overall shape. Small: borders follow fine processes.
Min signal	histogram · 0 (off)	Pixels whose smoothed signal is below this are excluded from every domain: background, gaps, vessels.
Min domain area (µm²)	0–5000 · 0 (off)	Domains smaller than this lose their seed and are re-flooded by their neighbours instead of surviving as slivers. Repeats up to five times.

Max domain area (µm²)	0–100 000 · 0 (off)	Each domain is trimmed to this area, keeping the pixels nearest its seed. Useful at tissue edges where a lone seed would otherwise claim a huge region.



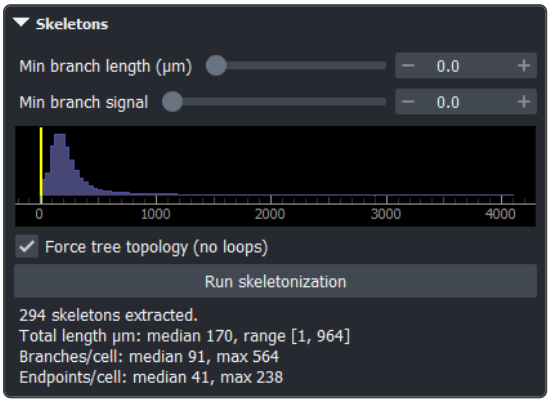
The whole section after tessellation. Domain colours are arbitrary; to check one domain, hover over it: napari's status bar shows its label number, which is the `domain_id` in the CSV.

PER-CELL MORPHOLOGY

8 Skeletons and Sholl analysis

Because each cell already has its own domain, tracing it is simple: skeletonize inside the domain, root the tree at the seed, count.

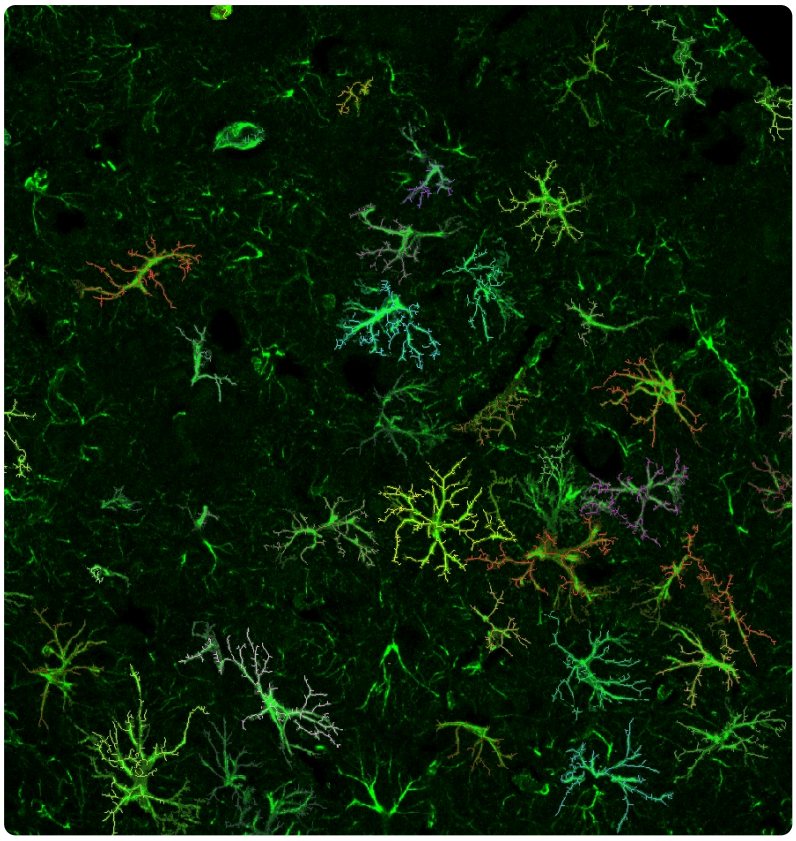
Skeletons



Run skeletonization thins each domain to a one-pixel-wide skeleton (scikit-image), keeps its largest connected piece, roots it at the seed, and decomposes it into branches with `skan`. The **Skeletons** layer shows the result in each domain's colour.

Skeletons follow the domain mask, so they describe the cell only if the domains were carved with *Min signal* (chapter 7). On uncarved territories the skeleton traces the territory's shape instead.

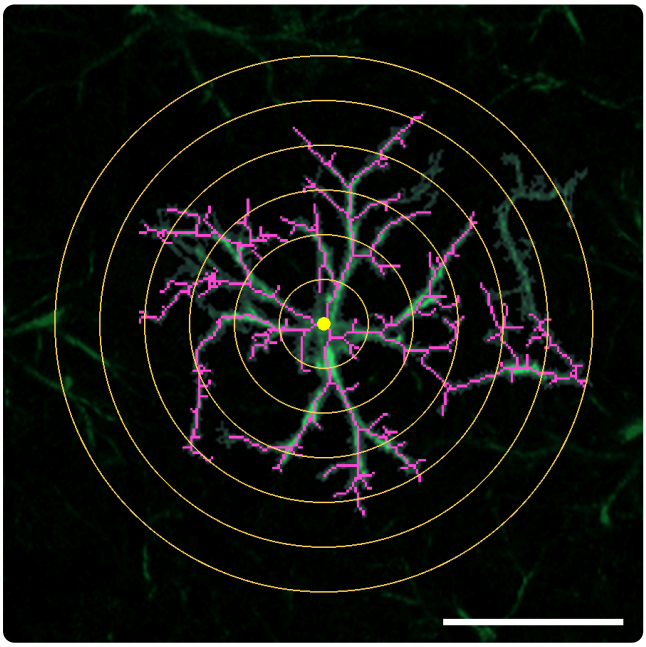
CONTROL	RANGE · DEFAULT	WHAT IT DOES
Min branch length (µm)	0–20 · 0 (off)	Prunes end twigs shorter than this…
Min branch signal	histogram · 0 (off)	…and end twigs whose mean signal is below this. A twig is pruned only if it fails **every** test that is switched on: a short twig survives if bright, a dim one if long. Pruning repeats until nothing changes.
Force tree topology	on	Breaks loops in the skeleton (common where processes cross), keeping the brightest paths and cutting the dimmest one in each loop, so every cell is a tree and branch counts are well defined.



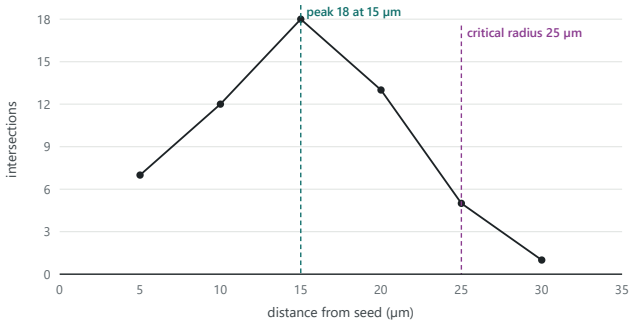
Skeletons over the signal, domains faint underneath.

Sholl analysis

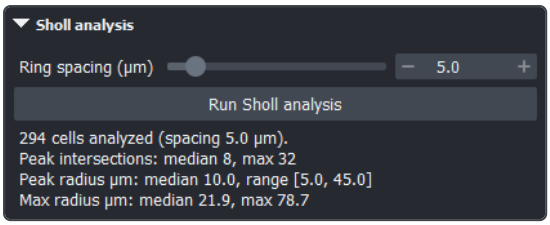
Sholl analysis counts how many times a cell's arbor crosses concentric circles centred on the soma. DOMINION counts *branches*, not pixels: within each annulus, each connected piece of skeleton is one intersection, so a long radial branch counts once per ring.



One cell. Domain (grey), skeleton (magenta), seed (yellow) and rings every 5 µm. Scale bar 20 µm.



Its profile: intersections at each ring radius, with the two radii the CSV reports.



Ring spacing (µm) (0.5–50, default 5) sets the distance between rings. Finer spacing gives a smoother profile but each ring counts fewer pixels; 2–5 µm suits astrocytes at typical confocal resolution.

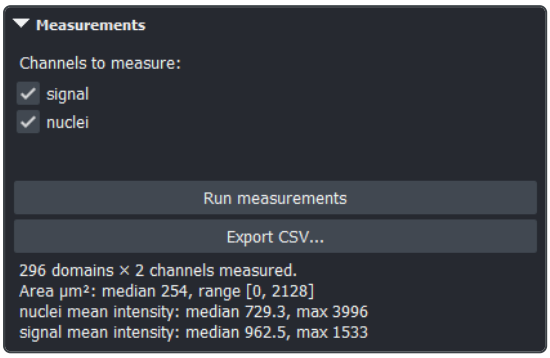
The summary under the button gives medians across cells; the per-cell numbers go into the CSV.

METRIC	MEANING
`peak_intersections`	most crossings at any ring: how bushy the arbor is at its widest
`peak_radius_um`	the radius where that peak occurs
`max_radius_um`	distance from the seed to the farthest skeleton pixel: the arbor's reach
`critical_radius_um`	first radius beyond the peak where crossings fall to half the peak or fewer
`auc`	sum of crossings over all rings, a proxy for total branch length
`ramification_index`	peak crossings ÷ crossings at the first ring: how much the arbor branches out from the primary processes

OUTPUTS

9 Measurements and export

One row per domain, every channel measured inside it, plus skeleton and Sholl columns if you ran them.



Tick the channels to measure (signal, nuclei and any extra channels) and click **Run measurements**. The summary shows the domain count and median area and intensity.

Export CSV... asks where to save, defaulting to the image's folder, and writes two files:

- `image_domains.csv`: one row per domain;
- `image_domains_summary.csv`: one `metric,value` row per image-level number, including every parameter used, so the file documents how it was made.

Run Skeletons and Sholl *before* exporting to get their columns; otherwise those cells are left empty.

Per-domain table image_domains.csv

COLUMNS	MEANING
`domain_id`	the domain's label in the Domains layer
`area_um2`	domain area
`centroid_y_px, centroid_x_px`	domain centroid, in pixels (multiply by the pixel size for µm)
`eccentricity, solidity`	shape: 0 = circle … 1 = line; area ÷ convex-hull area
`perimeter_um, equivalent_diameter_um`	outline length; diameter of a circle of the same area
`axis_major_um, axis_minor_um`	axes of the best-fit ellipse
`<channel>_mean, _median, _sum, _min, _max, _std`	intensity statistics of each measured channel inside the domain (`signal_`, `nuclei_`, `channel_2_` …)
`skel_total_length_um, skel_n_branches, skel_n_endpoints, skel_n_branchpoints`	from Skeletons
`sholl_*`	the six Sholl metrics (chapter 8)

Image summary image_domains_summary.csv

Image size and pixel size; tissue area; `n_domains` and `domain_density_per_mm2`; domain area mean, median, SD, quartiles, min and max; median and mean of each channel's per-domain mean intensity; medians of the skeleton and Sholl metrics; then every parameter, prefixed by its stage (`seeds__sigma_um`, `tessellation__min_signal`, …).

Answering common questions

QUESTION	WHERE THE ANSWER IS
How many objects?	`n_domains`, or `domain_density_per_mm2` to compare sections of different sizes. For counts per region, draw an ROI per region and export each.

How much of marker X per cell?	Load a multi-channel TIFF, tick X in Measurements, use `X_mean` (concentration) or `X_sum` (total amount) per domain.
How big, how branched?	`area_um2` (territory), skeleton and Sholl columns (arbor).
Who neighbours whom?	Not computed in the app yet. Use the centroids, as below.

Nearest-neighbour distances from the exported centroids (Python with pandas and SciPy, both already in the `dominion` environment):

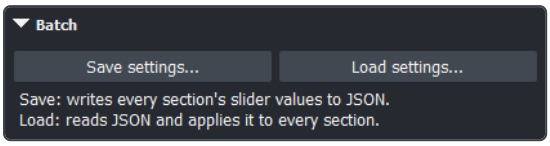
```
import pandas as pd
from scipy.spatial import cKDTree
df  = pd.read_csv(r"C:\path\to\image_domains.csv")
px  = 0.2857                                  # µm per pixel: pixel_size_um in the summary CSV
xy  = df[["centroid_x_px", "centroid_y_px"]].to_numpy() * px
d, _ = cKDTree(xy).query(xy, k=2)             # k=2: the nearest point is the cell itself
df["nn_distance_um"] = d[:, 1]
```

For true neighbour relationships (which domains share a border), use the Domains layer: in napari's console, `viewer.layers["Domains"].data` is the label image, and pixels where two different labels touch are the shared borders.

REPRODUCIBILITY

10 Saving settings

Tune once, then apply the same parameters to every image in the experiment.



`Save settings...` writes every section's controls (method, sliders, check boxes, the measured channels) to a JSON file, by default `image_settings.json` next to the image.

`Load settings...` reads one back and sets every control. It does **not** run anything: click each Run button in turn, top to bottom.

A settings file is plain JSON that you can read, keep with your data, or cite in a methods section:

```
{ "sections": {
    "signal_seed_finding": { "method": "Local maxima of smoothed signal", "sigma_um": 6.6,
                             "signal_threshold": 266.0, "min_distance_um": 15.0, "peak_strength": 338.8
},
    "tessellation":        { "signal_influence": 0.8, "smoothing_sigma_um": 0.1, "min_signal": 315.0,
                             "min_area_um2": 0.0, "max_area_um2": 0.0 },
    "skeletons":           { "min_branch_length_um": 0.0, "min_branch_signal": 0.0, "force_tree": true
},
    "sholl_analysis":      { "ring_spacing_um": 5.0 },
    "measurements":        { "channels_checked": ["signal", "nuclei"] } } }
```

These are the settings behind the screenshots in this manual. The thresholds are raw intensity values, so they carry over only between images acquired and exported the same way. Keep a settings file per staining batch.

FOR A WHOLE EXPERIMENT

- Tune on two or three representative images (or ROIs of them), then save.
- For each image: open it, `Load settings...`, Run each section, `Export CSV...`. Each export carries its parameters in the summary file.
- Combine the `*_domains.csv` files afterwards (for example with `pandas.concat`), adding a column for the image name.
- There is no unattended run-a-folder mode yet; it is planned.

PRACTICE

11 Choosing parameters

Tune from upstream to downstream, judge each stage by eye before moving on, and change one thing at a time.

A tuning order that converges

1. **Check the input.** Is the pixel size right (hover over the image: napari shows coordinates in µm)? Is the background exactly zero? Is the right channel the signal?
2. **Pick a test region.** Draw an ROI over a representative patch with dense and sparse areas. Every Run is then quick.
3. **Seeds first, and only seeds.** Zoom in and compare dots with cells. Aim for one seed per soma. Fix *merges* (two cells, one seed) by lowering σ or Min peak distance; fix *splits* (one cell, several seeds) by raising them. Then use the live Min peak intensity to drop seeds on background.
4. **Tessellation.** Start at Signal influence 0.8, smoothing 1–2 µm, Min signal 0. Check that borders run between cells, not through them. Then decide whether you want territories (keep Min signal 0) or carved cells (raise Min signal until the domains hug the stain).
5. **Clean-up.** Use Min domain area for slivers, Max domain area for edge seeds that claim too much.
6. **Save settings, clear the ROI, load settings, run on the whole image.**

Rules of thumb astrocytes, GFAP, confocal 20–40×

PARAMETER	START AT	WHY
Seed σ	⅓ soma diameter	Smooths processes away so each soma gives one peak.
Min peak distance	0.6–0.8 × cell spacing	Astrocyte somata are rarely closer than ~15 µm.
Signal influence	0.8	Strong enough to bend borders around processes, not so strong that bright vessels capture territory.
Min signal (carving)	just above the background peak	The histogram's tall left peak is background; the long tail is cells.
Sholl spacing	2–5 µm	Several rings per process segment without counting pixel noise.

Comparing groups

- **Keep parameters fixed across conditions.** Tune on a mix of control and treated images, then use one settings file for all. If a threshold must differ, it is a confound: report it.
- **Watch diffuse signal.** Reactive astrocytes with broad GFAP and no clear soma peak are undercounted by local maxima; try the distance-transform method or nuclei mode, and check seed counts against a manual count on a few fields.
- **There is no built-in ground truth.** Validate on a handful of hand-annotated fields before trusting group differences.

WHEN SOMETHING LOOKS WRONG

12 Troubleshooting

Grouped by when it happens.

Installing and starting

napari fails to start with `QtBindingsNotFoundError` or a Qt DLL error.

> PyQt6 does not load on some Windows machines. The easy installer already switches to PyQt5; on a manual install run the two `pip` lines in chapter 2.

"DOMINION — Full version required" when starting nuclei mode.

> This is a Basic install. Run **Install DOMINION (Full).bat**, or the Full manual install, then try again.

The installer window closes or stops with an error.

> Run it again and read the last lines before pressing a key. A failed download (proxy, firewall) of Miniforge or packages is the usual cause; installing Miniforge yourself first also works, since the installer uses any conda it finds.

The Desktop icon does nothing, or says DOMINION isn't installed.

> The icon starts `installers\Launch DOMINION.bat` in the folder you installed from. If that folder was moved or deleted, run the installer again from its new place.

StarDist takes minutes.

> TensorFlow is running on the CPU. That is expected on a CPU-only Full install. For GPU use you need an NVIDIA GPU and the GPU Full install. Results are cached, so re-runs are instant.

Loading an image

A warning that the pixel size is missing, or distances that make no sense.

> The TIFF has no resolution metadata. Set it in Fiji (**Image ▸ Properties**) and save as TIFF again.

"Expected a 2D single-channel TIFF or a 3D CYX TIFF".

> The file is a Z-stack, time series, or has channels last. Save a single plane (or a projection) with channels first.

Domains spill into the background.

> The background is not exactly zero. Clear it before loading (chapter 4), or raise *Min signal* in Tessellation.

Results

A Run button is greyed out.

> Its input does not exist yet: run the section above it. Changing channels or the ROI clears every result.

Too many seeds, several per cell.

> Raise Smoothing σ and Min peak distance; raise Min peak intensity.

Too few seeds, neighbouring cells share one.

> Lower σ and Min peak distance; lower the signal threshold. For diffuse cells, try the distance-transform method.

Thin slivers between real domains.

> Spurious seeds at cell edges. Raise Min peak intensity, or set Min domain area.

Domain borders cut through cells.

> Raise Signal influence, or lower the tessellation smoothing so borders follow processes.

After loading settings nothing changed on screen.

> Loading only sets the controls. Click each Run button, top to bottom.

Skeletons look like blobs or territory outlines.

> The domains are uncarved territories. Raise Min signal in Tessellation, re-run, then re-run Skeletons.

Skeleton or Sholl columns are empty in the CSV.

> Run Skeletons and Sholl before Export, and after the latest tessellation.

Skeletonization is slow on a big image.

> It scales with the number of cells: about 25 s for 300 cells on the example image. Tune on an ROI.

EVERY CONTROL

13 Parameter reference

Live = takes effect at once. Run = takes effect on the section's next Run.

SECTION · CONTROL	RANGE · DEFAULT	WHEN	EFFECT
Channels · Signal / Nuclei	channel 0 / 1	at once	Which channel defines objects / nuclei. Clears all results.
ROI · Apply / Clear	whole tissue	at once	Restrict to the last drawn shape. Clears all results.
Seeds · Method	local maxima	Run	Local maxima of smoothed signal, or distance-transform peaks.
Seeds · Smoothing σ	0–20 µm · 5	Run	Blur before peak finding.
Seeds · Signal threshold	data · median	Run	Ignore pixels below.
Seeds · Min peak distance	1–100 µm · 15	Run	Minimum seed spacing.
Seeds · Min peak intensity/depth	data · 0	live	Drop weak seeds.
Nuclei · prob_thresh	0–1 · 0.50	Run	StarDist detection confidence.
Nuclei · nms_thresh	0–1 · 0.40	Run	StarDist overlap allowed.
Classify · T	data · median	Run	Signal a pixel needs to count toward a score.
Classify · R	1–50 µm · 10	Run	Scoring radius around each nucleus.
Classify · α	0–1 · 0.50	Run	Intensity (1) vs proximity (0) weighting.
Classify · θ	data · median	live	Keep nuclei scoring at least θ.
Tessellation · Signal influence	0–1 · 0.5	Run	Signal vs distance in the watershed.
Tessellation · Smoothing σ	0–10 µm · 2	Run	Blur of the guiding signal.
Tessellation · Min signal	data · 0 (off)	Run	Exclude dim pixels: carve domains to the cell.
Tessellation · Min domain area	0–5000 µm² · 0	Run	Absorb small domains into neighbours.
Tessellation · Max domain area	0–100k µm² · 0	Run	Trim domains to the pixels nearest their seed.
Skeletons · Min branch length	0–20 µm · 0	Run	Prune short end twigs (if also dim).
Skeletons · Min branch signal	data · 0	Run	Prune dim end twigs (if also short).
Skeletons · Force tree	on	Run	Cut the dimmest path in each loop.
Sholl · Ring spacing	0.5–50 µm · 5	Run	Distance between rings.
Measurements · channels	all ticked	Run	Which channels get intensity columns.